# THE TOTAL LOCALIZED CHERN CLASS OF A DIRECT SUM IS THE PRODUCT OF THE TOTAL CLASSES OF THE SUMMANDS

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

*Editorial note.* The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

*Editorial note.* Canonical location of the counted proposition: Stacks tag 0FAY (sources/stacks-chow/pinned-original-source.tex line 10499).

*Allowed standard prerequisites (editorial).*

- The localized Chern classes and the truncated total class c^{(p)}(Q). Same chapter, Lemma labelled lemma-localized-chern-pre, cited at proof lines 10519-10521 and 10536-10540, and Remark labelled remark-ring-loc-classes, invoked in the statement. Fixes the classes being multiplied and the reduction of c'_i(Q) to c_i of the restriction to the fiber over zero.
- Additivity of the localized Chern classes. Same chapter, Lemma labelled lemma-additivity-on-perfect, cited at proof lines 10525-10527. Handles the degrees below p + p'.
- The product formula over E and the description of c'_p via pushforward and C. Same chapter, Lemmas labelled lemma-silly-sum-c and lemma-silly, cited at proof lines 10532-10534 and 10565-10567, and Lemma labelled lemma-localized-chern-pre for the formula involving C. Lets the computation be carried out on the closed subscheme E inside W_infinity.
- Commutation of bivariant Chern classes with capping and the homomorphism lemma. Same chapter, Lemma labelled lemma-cap-commutative-chern, cited at proof lines 10592-10594, and Lemma labelled lemma-homomorphism, cited at proof lines 10575-10578 and 10586-10589. Settles the two mixed cases where one index is below the truncation bound.

*Source reference.* Original lemma 0F9K: sources/stacks-chow/pinned-original-source.tex, source lines 10204–10204. Same-chapter pointer into the author environment lemma at source line 10204; invoked at proof line 10522 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

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*Date*: stacks-project:chow.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Source reference.* Original reference 0FA0: sources/stacks-chow/pinned-original-source.tex, source lines 6419–6419. Same-chapter pointer into the author environment remark at source line 6419; invoked at proof line 10547 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

*Source reference.* Original lemma 0F9F: sources/stacks-chow/pinned-original-source.tex, source lines 9427–9427. Same-chapter pointer into the author environment lemma at source line 9427; invoked at proof line 10524 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

*Source reference.* Original lemma 0FAL: sources/stacks-chow/pinned-original-source.tex, source lines 9889–9889. Same-chapter pointer into the author environment lemma at source line 9889; invoked at proof line 10535 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

*Source reference.* Original reference 0FA1: sources/stacks-chow/pinned-original-source.tex, source lines 6451–6451. Same-chapter pointer into the author environment remark at source line 6451; invoked at proof line 10554 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

*Source reference.* Original lemma 0F9H: sources/stacks-chow/pinned-original-source.tex, source lines 9695–9695. Same-chapter pointer into the author environment lemma at source line 9695; invoked at proof line 10561 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

*Source reference.* Original lemma 0FAV: sources/stacks-chow/pinned-original-source.tex, source lines 10353–10353. Same-chapter pointer into the author environment lemma at source line 10353; invoked at proof line 10571 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

*Source reference.* Original lemma 02UA: sources/stacks-chow/pinned-original-source.tex, source lines 7581–7581. Same-chapter pointer into the author environment lemma at source line 7581; invoked at proof line 10594 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

**Lemma 0.1.** *In Lemma 0F9K assume $Q|_T$ is isomorphic to a finite locally free $\mathcal{O}_T$-module of rank $< p$. Assume we have another perfect object $Q' \in D(\mathcal{O}_W)$ whose Chern classes are defined with $Q'|_T$ isomorphic to a finite locally free $\mathcal{O}_T$-module of rank $< p'$ placed in cohomological degree 0. With notation as in Remark 0FA0 set*

$$c^{(p)}(Q) = 1 + c_1(Q|_{X\times\{0\}}) + \ldots + c_{p-1}(Q|_{X\times\{0\}}) + c'_p(Q) + c'_{p+1}(Q) + \ldots$$

*in $A^{(p)}(Z \to X)$ with $c'_i(Q)$ for $i \geq p$ as in Lemma 0F9K. Similarly for $c^{(p')}(Q')$ and $c^{(p+p')}(Q \oplus Q')$. Then $c^{(p+p')}(Q \oplus Q') = c^{(p)}(Q)c^{(p')}(Q')$ in $A^{(p+p')}(Z \to X)$.*

*Proof.* Recall that the image of $c'_i(Q)$ in $A^p(X)$ is equal to $c_i(Q|_{X\times\{0\}})$ for $i \geq p$ and similarly for $Q'$ and $Q \oplus Q'$, see Lemma 0F9K. Hence the equality in degrees $< p + p'$ follows from the additivity of Lemma 0F9F.

Let's take $n \geq p + p'$. As in the proof of Lemma 0F9K let $E \subset W_\infty$ denote the inverse image of $Z$. Observe that we have the equality

$$c^{(p+p')}(Q|_E \oplus Q'|_E) = c^{(p)}(Q|_E)c^{(p')}(Q'|_E)$$

in $A^{(p+p')}(E \to W_\infty)$ by Lemma 0FAL. Since by construction

$$c'_p(Q \oplus Q') = (E \to Z)_* \circ c'_p(Q|_E \oplus Q'|_E) \circ C$$

we conclude that suffices to show for all $i + j = n$ we have

$$(E \to Z)_* \circ c_i^{(p)}(Q|_E) c_j^{(p')}(Q'|_E) \circ C = c_i^{(p)}(Q) c_j^{(p')}(Q')$$

in $A^n(Z \to X)$ where the multiplication is the one from Remark 0FA0 on both sides. There are three cases, depending on whether $i \geq p$, $j \geq p'$, or both.

Assume $i \geq p$ and $j \geq p'$. In this case the products are defined by inserting $(E \to W_\infty)_*$, resp. $(Z \to X)_*$ in between the two factors and taking compositions as bivariant classes, see Remark 0FA1. In other words, we have to show

$$(E \to Z)_* \circ c'_i(Q|_E) \circ (E \to W_\infty)_* \circ c'_j(Q'|_E) \circ C = c'_i(Q) \circ (Z \to X)_* \circ c'_j(Q')$$

By Lemma 0F9H the left hand side is equal to

$$(E \to Z)_* \circ c'_i(Q|_E) \circ c_j(Q'|_{W_\infty}) \circ C$$

Since $c'_i(Q) = (E \to Z)_* \circ c'_i(Q|_E) \circ C$ the right hand side is equal to

$$(E \to Z)_* \circ c'_i(Q|_E) \circ C \circ (Z \to X)_* \circ c'_j(Q')$$

which is immediately seen to be equal to the above by Lemma 0FAV.

Assume $i \geq p$ and $j < p$. Unwinding the products in this case we have to show

$$(E \to Z)_* \circ c'_i(Q|_E) \circ c_j(Q'|_{W_\infty}) \circ C = c'_i(Q) \circ c_j(Q'|_{X \times \{0\}})$$

Again using that $c'_i(Q) = (E \to Z)_* \circ c'_i(Q|_E) \circ C$ we see that it suffices to show $c_j(Q'|_{W_\infty}) \circ C = C \circ c_j(Q'|_{X \times \{0\}})$ which is part of Lemma 0FAV.

Assume $i < p$ and $j \geq p'$. Unwinding the products in this case we have to show

$$(E \to Z)_* \circ c_i(Q|_E) \circ c'_j(Q'|_E) \circ C = c_i(Q|_{Z \times \{0\}}) \circ c'_j(Q')$$

However, since $c'_j(Q|_E)$ and $c'_j(Q')$ are bivariant classes, they commute with capping with Chern classes (Lemma 02UA). Hence it suffices to prove

$$(E \to Z)_* \circ c'_j(Q'|_E) \circ c_i(Q|_{W_\infty}) \circ C = c'_j(Q') \circ c_i(Q|_{X \times \{0\}})$$

which we reduces us to the case discussed in the preceding paragraph. □